

CNT 4714 – Project 1 – Fall 2024

Title: “Program Assignment 1: Event-Driven Programming”

Points: 100 points

Due Date: **Sunday September 8, 2024 by 11:59 pm (WebCourses time)**

Objectives: To practice event-driven programming using Java-based GUIs. To refresh your basic Java skills. To simulate (albeit at a very high-level) an enterprise application.

Description: Develop a Java application that creates a standalone GUI application that simulates an e-store. We’ll call our store Nile Dot Com... we’re not quite as big as Amazon.com!¹ The application will allow the user to place in stock items into a shopping cart and once all items are included, total all costs (including tax), produce an invoice, and append a transaction log file.

Your program development must include the following steps:

1. Create a main GUI containing the following components:
 - a. An area that allows the user to input data into the application along with the descriptive text that describes each input area.
 - b. An area that shows the current contents of the shopping cart.
 - c. A total of six buttons as shown below with functionality as described below. The various input fields and buttons on the interface are only accessible at certain points during a user’s interaction with the e-store.
2. An input file named “`inventory.csv`”. This is a comma separated file which contains the data that will be read by the application when the user makes a selection. Each line in this file contains five entries: an item id (a string), a quoted string containing the description of the item, an in stock status (a string), the quantity on hand (an integer), and the unit price for one of the item (a double). A sample file is provided for you on WebCourses. Feel free to create your own input file for testing purposes or augment the provided input file.
3. An output file (append only) named “`transactions.csv`” must be created that uniquely identifies and logs each user transaction with the e-store. The unique transaction id will be generated as a permutation of the date/time string when the transaction occurred (see below).

Restrictions:

- Your source file shall begin with comments containing the following information:

```
/* Name: <your name goes here>  
   Course: CNT 4714 – Fall 2024  
   Assignment title: Project 1 – An Event-driven Enterprise Simulation  
   Date: Sunday September 8, 2024  
*/
```
- We’ll assume that a shopping cart can hold at most 5 separate items. You don’t need to worry about handling more than 5 separate items/transaction.
- We’ll assume that our client’s know the sequence of events that need to occur. This will reduce the amount of error detection/correction that you need to implement (see more below).
- Assume the tax rate is 6%. Discount rates as shown below on page X.

¹ Source = <https://www.wonderslist.com/top-10-largest-rivers/>

Input Specification: The file "inventory.csv" as described above (see example below as well).

Output Specification: Output is to appear in the specified components of the GUI and various message boxes that appear, plus the contents of the "transactions.csv" log file that will be generated.

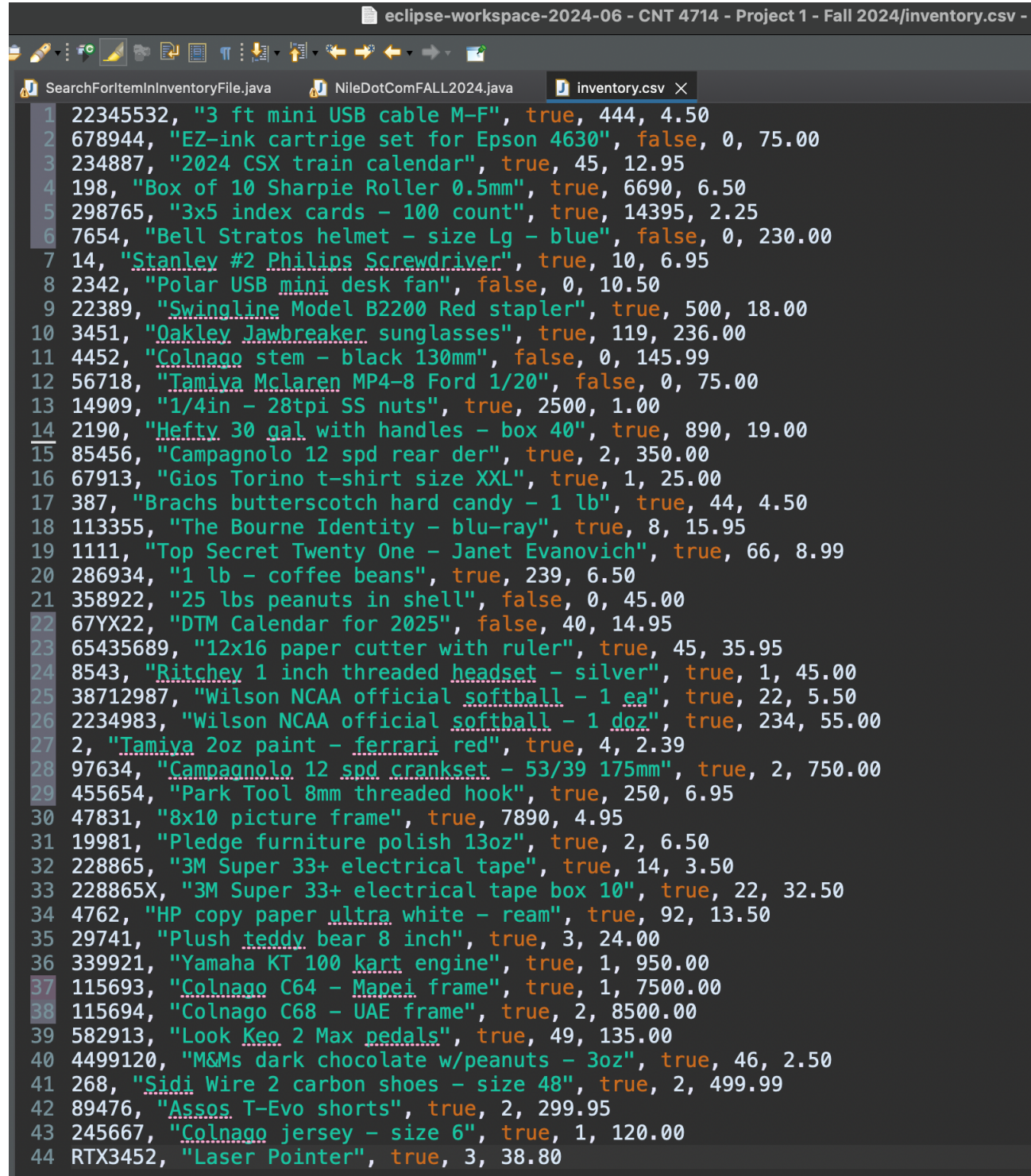
Deliverables:

Submit all of the following via WebCourses no later than 11:59pm Sunday September 8, 2024. Zip everything into a single folder.

- (1) A working copy of your source code (all .java files) and any necessary supporting libraries.
- (2) Include a file that contains (**at least**) the following nine screen shots. All of these are taken from the perspective of your executing application (i.e., from your GUI).
 - a. The initial GUI instance (like number 2 in Additional Information below).
 - b. The GUI after the user has entered an item number (ID) and quantity and clicked the "Search For Item #X" button (like number 4 below).
 - c. The GUI after the user has added an item from screenshot (b) above but before searching for the next item (like numbers 7 and 9 below).
 - d. The GUI after the user has searched for the next item after the one in (b) above was added to the cart, but before adding this new item to the cart (like number 8 below).
 - e. The GUI and the message dialog box when the user clicks the "View Current Order" button with multiple items (at least two) in the order (like number 10 below).
 - f. The message dialog box (the invoice) when the user clicks the "Check Out" button for an order containing multiple items (like number 11 below).
 - g. The GUI and the message dialog box when the user has entered an item number (ID) and quantity where the quantity is greater than the number in stock (like number 16 below).
 - h. The GUI and the message dialog box when the item the user has selected is out of stock (like number 13 below).
 - i. The GUI and the message dialog box when the item the user has selected is not contained in the inventory file (like 6 below).
 - You may include more screenshots than the nine listed above if you think they will illustrate additional features of your application than those already listed above.
- (2) Include your "transactions.csv" file as it stands after several different transactions have been recorded. You should have at least **5** different transactions in the file spanning a time period of at least **two hours** with at least **three** of these transactions having **four or five** items in the transaction. (similar to the one shown on the last page of this document).

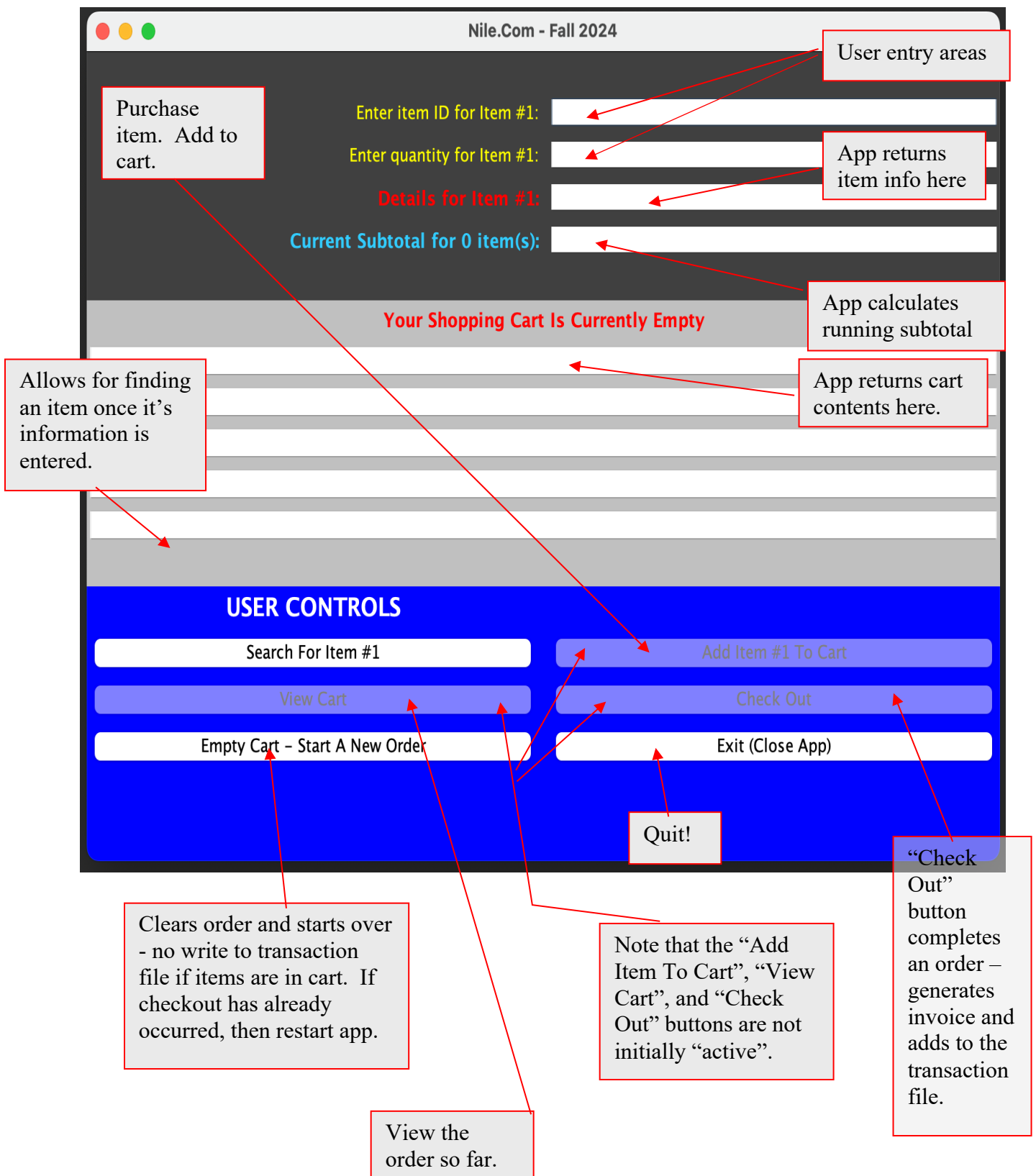
Additional Information:

1. Screen shot of the contents of an example `inventory.csv` file.



```
eclipse-workspace-2024-06 - CNT 4714 - Project 1 - Fall 2024/inventory.csv -  
SearchForItemInInventoryFile.java NileDotComFALL2024.java inventory.csv X  
1 22345532, "3 ft mini USB cable M-F", true, 444, 4.50  
2 678944, "EZ-ink cartridge set for Epson 4630", false, 0, 75.00  
3 234887, "2024 CSX train calendar", true, 45, 12.95  
4 198, "Box of 10 Sharpie Roller 0.5mm", true, 6690, 6.50  
5 298765, "3x5 index cards - 100 count", true, 14395, 2.25  
6 7654, "Bell Stratos helmet - size Lg - blue", false, 0, 230.00  
7 14, "Stanley #2 Philips Screwdriver", true, 10, 6.95  
8 2342, "Polar USB mini desk fan", false, 0, 10.50  
9 22389, "Swingline Model B2200 Red stapler", true, 500, 18.00  
10 3451, "Oakley Jawbreaker sunglasses", true, 119, 236.00  
11 4452, "Colnago stem - black 130mm", false, 0, 145.99  
12 56718, "Tamiya McLaren MP4-8 Ford 1/20", false, 0, 75.00  
13 14909, "1/4in - 28tpi SS nuts", true, 2500, 1.00  
14 2190, "Hefty 30 gal with handles - box 40", true, 890, 19.00  
15 85456, "Campagnolo 12 spd rear der", true, 2, 350.00  
16 67913, "Gios Torino t-shirt size XXL", true, 1, 25.00  
17 387, "Brachs butterscotch hard candy - 1 lb", true, 44, 4.50  
18 113355, "The Bourne Identity - blu-ray", true, 8, 15.95  
19 1111, "Top Secret Twenty One - Janet Evanovich", true, 66, 8.99  
20 286934, "1 lb - coffee beans", true, 239, 6.50  
21 358922, "25 lbs peanuts in shell", false, 0, 45.00  
22 67YX22, "DTM Calendar for 2025", false, 40, 14.95  
23 65435689, "12x16 paper cutter with ruler", true, 45, 35.95  
24 8543, "Ritchey 1 inch threaded headset - silver", true, 1, 45.00  
25 38712987, "Wilson NCAA official softball - 1 ea", true, 22, 5.50  
26 2234983, "Wilson NCAA official softball - 1 doz", true, 234, 55.00  
27 2, "Tamiya 2oz paint - ferrari red", true, 4, 2.39  
28 97634, "Campagnolo 12 spd crankset - 53/39 175mm", true, 2, 750.00  
29 455654, "Park Tool 8mm threaded hook", true, 250, 6.95  
30 47831, "8x10 picture frame", true, 7890, 4.95  
31 19981, "Pledge furniture polish 13oz", true, 2, 6.50  
32 228865, "3M Super 33+ electrical tape", true, 14, 3.50  
33 228865X, "3M Super 33+ electrical tape box 10", true, 22, 32.50  
34 4762, "HP copy paper ultra white - ream", true, 92, 13.50  
35 29741, "Plush teddy bear 8 inch", true, 3, 24.00  
36 339921, "Yamaha KT 100 kart engine", true, 1, 950.00  
37 115693, "Colnago C64 - Mapei frame", true, 1, 7500.00  
38 115694, "Colnago C68 - UAE frame", true, 2, 8500.00  
39 582913, "Look Keo 2 Max pedals", true, 49, 135.00  
40 4499120, "M&Ms dark chocolate w/peanuts - 3oz", true, 46, 2.50  
41 268, "Sidi Wire 2 carbon shoes - size 48", true, 2, 499.99  
42 89476, "Assos T-Evo shorts", true, 2, 299.95  
43 245667, "Colnago jersey - size 6", true, 1, 120.00  
44 RTX3452, "Laser Pointer", true, 3, 38.80
```

2. Initial GUI:



Notice the current status of the various labels, input fields, and output areas in the initial GUI and notice how they change over the following pages as the user interacts with the application.

3. GUI after user has entered first item number and the quantity of the item they want to purchase, but before clicking the “Search For Item #X” button.

The screenshot displays a web application window titled "Nile.Com - Fall 2024". The interface is divided into several sections. At the top, there are four input fields with labels: "Enter item ID for Item #1:" (containing "14"), "Enter quantity for Item #1:" (containing "1"), "Details for Item #1:" (empty), and "Current Subtotal for 0 item(s):" (empty). Below these fields is a message: "Your Shopping Cart Is Currently Empty". Underneath this message is a table with 7 rows and 2 columns, all of which are empty. At the bottom of the window is a blue section titled "USER CONTROLS" containing six buttons arranged in a 3x2 grid: "Search For Item #1", "Add Item #1 To Cart", "View Cart", "Check Out", "Empty Cart – Start A New Order", and "Exit (Close App)".

USER CONTROLS	
Search For Item #1	Add Item #1 To Cart
View Cart	Check Out
Empty Cart – Start A New Order	Exit (Close App)

4. GUI after user has selected the first item and clicked the “Search For Item #1” button.

The screenshot shows a web application window titled "Nile.Com - Fall 2024". The interface includes several input fields and buttons. The "Enter item ID for Item #1:" field contains the value "14". The "Enter quantity for Item #1:" field contains the value "1". The "Details for Item #1:" field displays "14 \"Stanley #2 Philips Screwdriver\" \$6.95 1 0% \$6.95". The "Current Subtotal for 0 item(s):" field is empty. Below these fields, a red message states "Your Shopping Cart Is Currently Empty". At the bottom, there are five buttons: "Search For Item #1", "View Cart", "Empty Cart - Start A New Order", "Add Item #1 To Cart", "Check Out", and "Exit (Close App)".

Enter item ID for Item #1: 14

Enter quantity for Item #1: 1

Details for Item #1: 14 "Stanley #2 Philips Screwdriver" \$6.95 1 0% \$6.95

Current Subtotal for 0 item(s):

Your Shopping Cart Is Currently Empty

Search For Item #1

View Cart

Empty Cart - Start A New Order

Add Item #1 To Cart

Check Out

Exit (Close App)

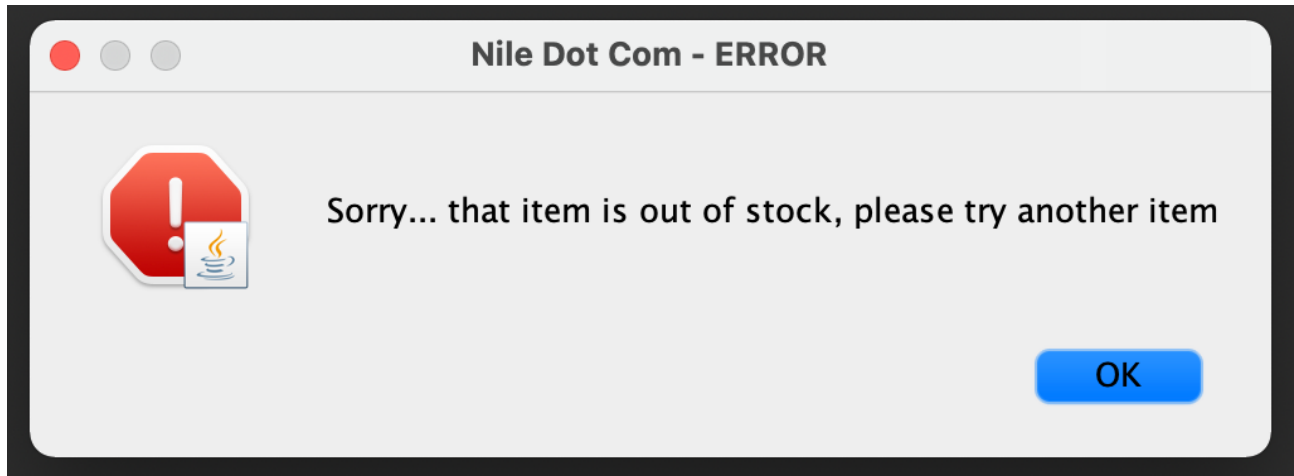
Information on the item selected/purchased has been extracted from the inventory file and written into the window on the line for item #1. Note proper currency formatting.

Discount is based on the quantity of an item. If:

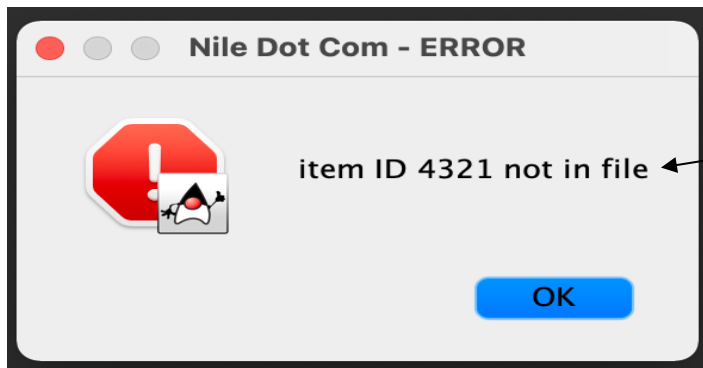
- 1-4 = 0%
- 5-9 = 10%
- 10-14 = 15%
- >=15 = 20%

Note that the “Add Item #1 To Cart” button is now active and the “Search For Item #X” button is inactive. The “View Cart” and “Check Out” buttons remain inactive.

5. When the user clicks on the “Search For Item #X” button and the item is found in the inventory, but the item is not in stock, then the following message would appear and the user is prompted to enter another product id. (see below as well)



6. When the user clicks on the “Search For Item #1” button, if the item is not found in the inventory, then the following message would appear on the user’s screen.



User entered an item ID that was not found in the inventory file (DB). Item is not processed and the user will need to enter a new item ID. Note that the item number will not increment in this case.

7. When the user clicks the “Add Item #1 To Cart” button, the GUI will be updated as shown:

The screenshot displays the Nile.Com - Fall 2024 application window. The interface is divided into several sections:

- Input Fields:** Two input fields at the top are labeled "Enter item ID for Item #2:" and "Enter quantity for Item #2:". A red box notes: "Label values change and input field cleared for next input." The "Enter item ID for Item #2:" field contains the text "14".
- Details for Item #1:** A section labeled "Details for Item #1:" displays "14 'Stanley #2 Philips Screwdriver' \$6.95 1 0% \$6.95".
- Subtotal:** A section labeled "Current Subtotal for 1 item(s):" displays "\$6.95". A red box notes: "Subtotal should reflect current total of all confirmed items."
- Shopping Cart:** A section titled "Your Shopping Cart Currently Contains 1 Item(s)" lists "Item 1 - SKU: 14, Desc: 'Stanley #2 Philips Screwdriver', Price Ea. \$6.95, Qty: 1, Total: \$6.95". A red box notes: "Modified version of the file line is added to shopping cart."
- User Controls:** A blue section at the bottom contains several buttons:
 - "Search For Item #2:" (with a red box note: "Still reflecting previous item.")
 - "Add Item #2 To Cart" (highlighted in blue)
 - "View Cart"
 - "Check Out"
 - "Empty Cart - Start A New Order"
 - "Exit (Close App)"A red box notes: "Note that the buttons change number." pointing to the "Add Item #2 To Cart" button.

8. User enters and finds (but has not yet added it to the cart) the next item in the order (which we'll make the last one for this order).

Nile.Com - Fall 2024

Enter item ID for Item #2:

339921

Enter quantity for Item #2:

1

Details for Item #1:

14 "Stanley #2 Philips Screwdriver" \$6.95 1 0% \$6.95

Current Subtotal for 1 item(s):

\$6.95

Your Shopping Cart Currently Contains 1 Item(s)

Item 1 - SKU: 14, Desc: "Stanley #2 Philips Screwdriver", Price Ea. \$6.95, Qty: 1, Total: \$6.95

USER CONTROLS

Search For Item #2

Add Item #2 To Cart

View Cart

Check Out

Empty Cart - Start A New Order

Exit (Close App)

9. The GUI after the user has entered the information for all the items and purchased the last item they wish to order.

Nile.Com - Fall 2024

Enter item ID for Item #3:

Enter quantity for Item #3:

Details for Item #2: 339921 "Yamaha KT 100 kart engine" \$950.00 1 0% \$950.00

Current Subtotal for 2 item(s): \$956.95

Your Shopping Cart Currently Contains 2 Item(s)

Item 1 - SKU: 14, Desc: "Stanley #2 Philips Screwdriver", Price Ea. \$6.95, Qty: 1, Total: \$6.95
Item 2 - SKU: 339921, Desc: "Yamaha KT 100 kart engine", Price Ea. \$950.00, Qty: 1, Total: \$950.00

USER CONTROLS

Search For Item #3	Add Item #3 To Cart
View Cart	Check Out
Empty Cart - Start A New Order	Exit (Close App)

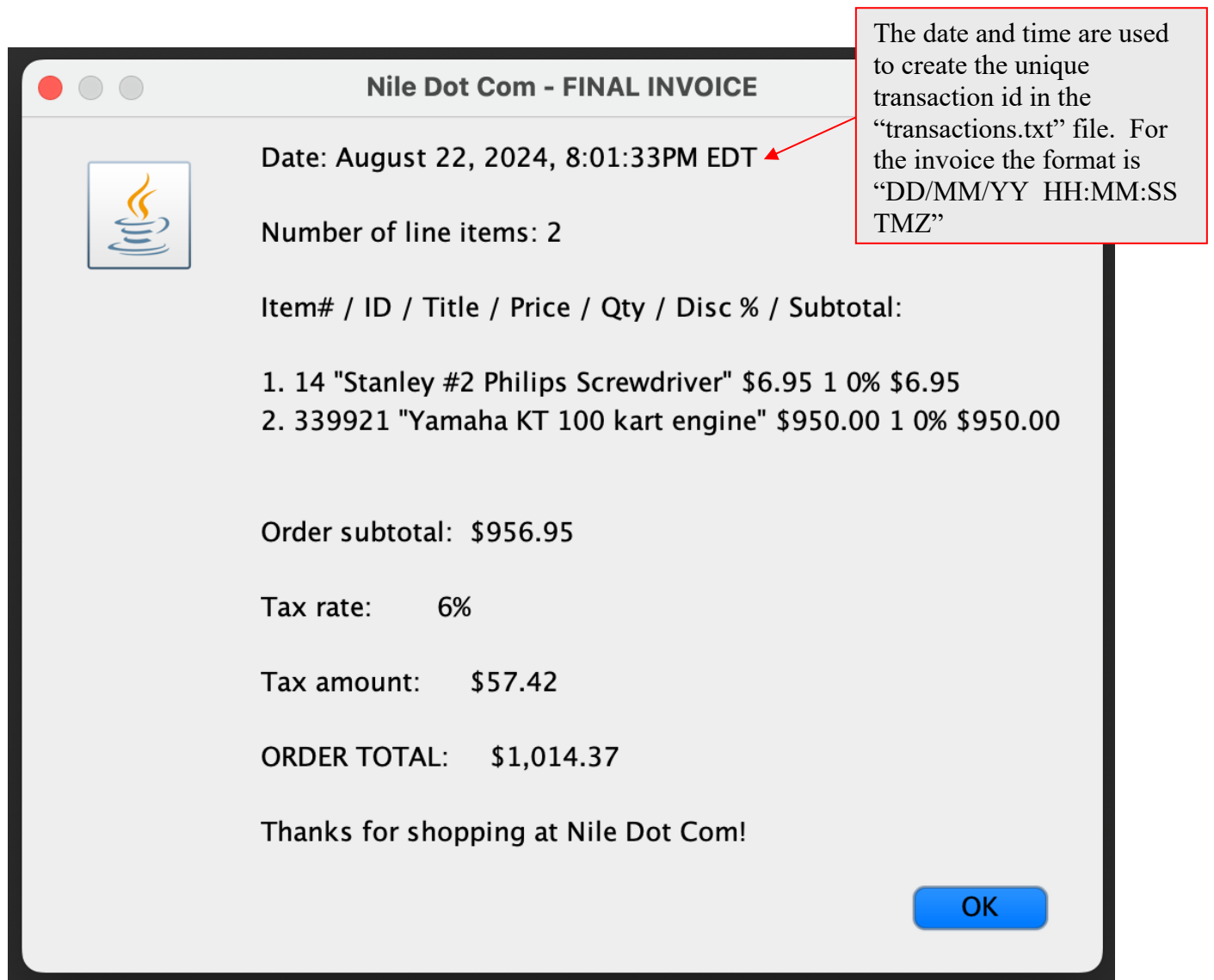
10. When the user clicks the “View Current Order” button, the following message box should appear, which is basically the same thing as occurred on each of the detail lines as the items were added to the cart by the user.

Nile Dot Com - Current Shopping Cart Status

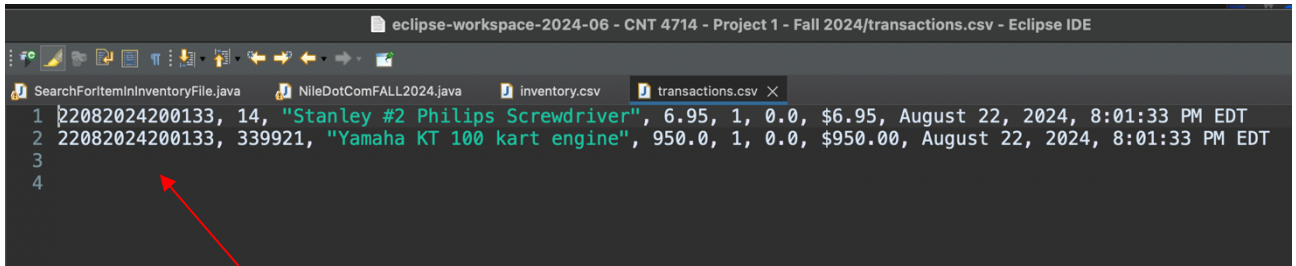


1. 14 "Stanley #2 Philips Screwdriver" \$6.95 1 0% \$6.95
2. 339921 "Yamaha KT 100 kart engine" \$950.00 1 0% \$950.00

11. When the user clicks the “Check Out” button, the final invoice message should be generated and displayed. In addition, the order details are added to the `transaction.csv` file.



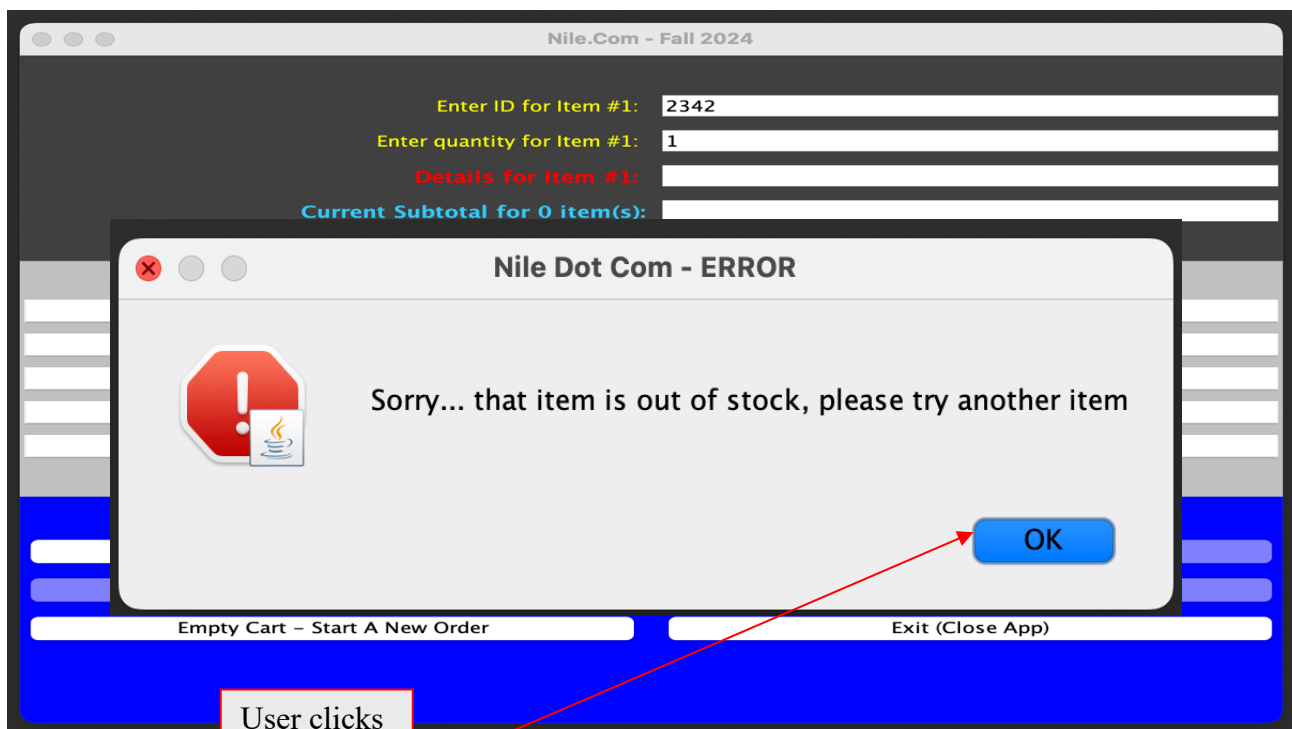
12. The transaction file after order shown above was finished. Note the unique transaction ids based on the date and time.



```
eclipse-workspace-2024-06 - CNT 4714 - Project 1 - Fall 2024/transactions.csv - Eclipse IDE
SearchForItemInInventoryFile.java NileDotComFALL2024.java inventory.csv transactions.csv
1 22082024200133, 14, "Stanley #2 Philips Screwdriver", 6.95, 1, 0.0, $6.95, August 22, 2024, 8:01:33 PM EDT
2 22082024200133, 339921, "Yamaha KT 100 kart engine", 950.0, 1, 0.0, $950.00, August 22, 2024, 8:01:33 PM EDT
3
4
```

The transaction occurred on 22/08/2024 at 20:01:33 (8:01:33pm). See the invoice message above. Permutation used here is: DDMMYYYYHHMMSS.

13. If the user enters an item number for an item that is not in stock (it is in the inventory – just not in stock – see number 5 above), the following message dialog should appear. Once the user clicks OK in the dialog box, then they would enter a new item number for another item and continue processing their current transaction. Notice below, that after clicking ok in the dialog box, the GUI is reset to the original item number and the input fields are empty.



User clicks "OK" in the dialog box.

Nile.Com - Fall 2024

Enter ID for Item #1:

Enter quantity for Item #1:

Details for Item #1:

Current Subtotal for 0 item(s):

Your Shopping Cart Is Currently Empty

Notice that after the “Not In Stock” error has occurred, the GUI has cleared the user entries in the two input fields and has not incremented the item count. User can now search again

USER CONTROLS

Search For Item #1

Add Item #1To Cart

View Cart

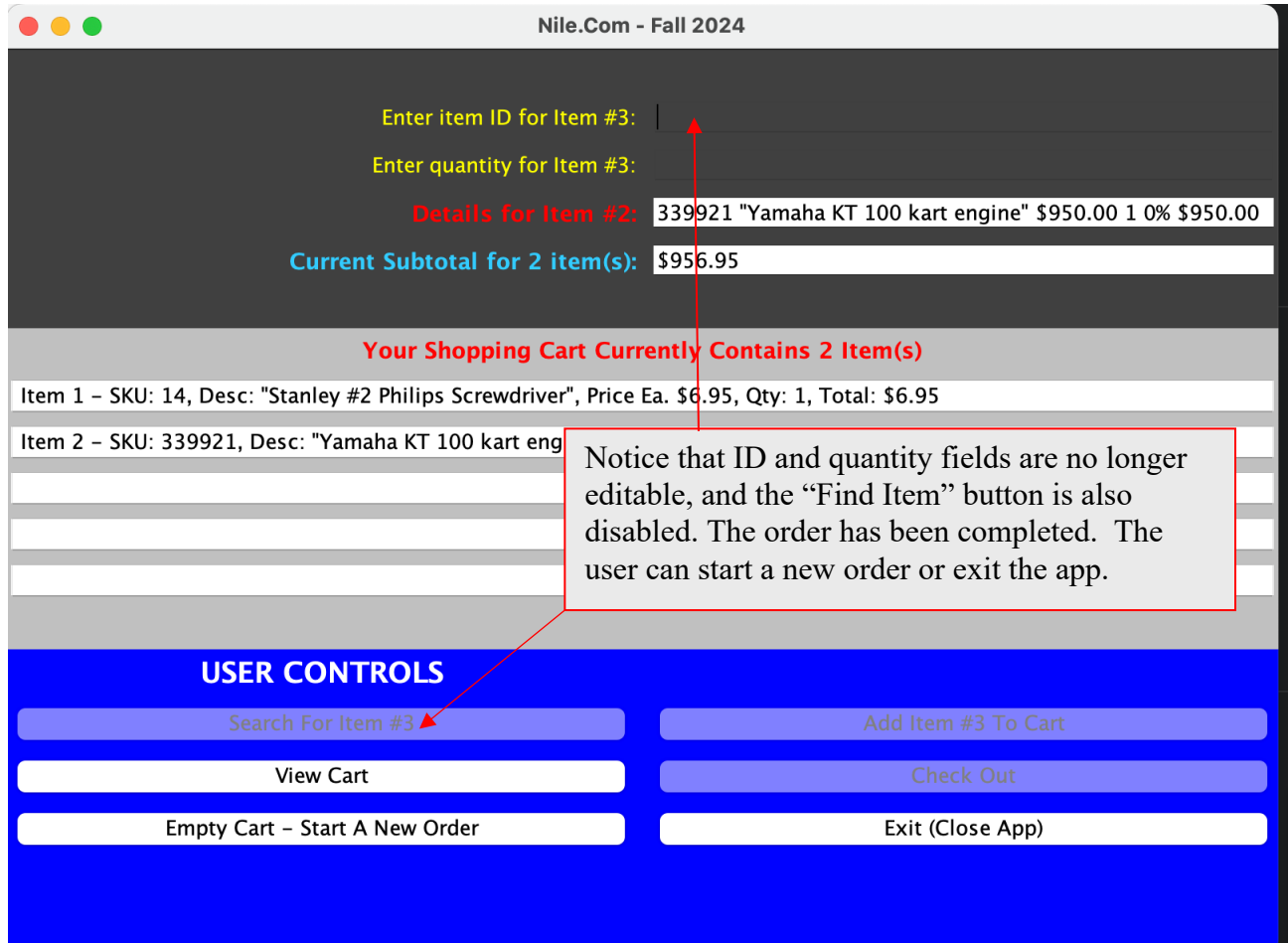
Check Out

Empty Cart – Start A New Order

Exit (Close App)

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14. The GUI after the user has clicked the “Check Out” button and also clicked “OK” on the invoice dialog box.



15. User has added 5 items to their shopping cart at this point. Cart is now full, so no more items can be added and the user must either check out or start a new order.

Nile.Com - Fall 2024

Enter item ID for Item #6:

Enter quantity for Item #6:

Details for Item #5: Campagnolo 12 spd crankset – 53/39 175mm" \$750.00 1 0% \$750.00

Current Subtotal for 5 item(s): \$860.51

Your Shopping Cart Currently Contains 5 Item(s)

Item 1 – SKU: 2190, Desc: "Hefty 30 gal with handles – box 40", Price Ea. \$19.00, Qty: 2, Total: \$38.00
Item 2 – SKU: 298765, Desc: "3x5 index cards – 100 count", Price Ea. \$2.25, Qty: 15, Total: \$27.00
Item 3 – SKU: 65435689, Desc: "12x16 paper cutter with ruler", Price Ea. \$35.95, Qty: 1, Total: \$35.95
Item 4 – SKU: 2, Desc: "Tamiya 2oz paint – ferrari red", Price Ea. \$2.39, Qty: 4, Total: \$9.56
Item 5 – SKU: 97634, Desc: "Campagnolo 12 spd crankset – 53/39 175mm", Price Ea. \$750.00, Qty: 1, Total: \$750.00

USER CONTROLS

Search For Item #5

Add Item #5 To Cart

View Cart

Check Out

Empty Cart – Start A New Order

Exit (Close App)

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16. The situation may also arise where there is not a sufficient quantity in stock of an item that the user would like to add to the cart. When this situation arises, you should print the following message to the user.

The screenshot shows a web application window titled "Nile.Com - Fall 2024". The main interface has a dark background with white text. It includes input fields for "Enter item ID for Item #2:" (containing "8543") and "Enter quantity for Item #2:" (containing "3"). Below these is a line for "Details for Item #1:" showing "Top Secret Twenty One – Janet Evanovich" with a price of \$8.99, a 20% discount, and a subtotal of \$17.98. A "Current Subtotal for 1 item(s):" field also shows \$17.98. An error dialog box is overlaid in the center, titled "Nile Dot Com - ERROR". It features a red octagonal icon with a white exclamation mark and a small coffee cup icon. The text inside the dialog reads: "Insufficient stock. Only 1 on hand. Please reduce the quantity." with an "OK" button at the bottom right. Below the dialog, a blue bar contains the text "USER CONTROLS". Underneath this bar are several buttons: "Search For Item #2", "Add Item #2 To Cart" (highlighted in blue), "View Cart", "Check Out", "Empty Cart – Start A New Order", and "Exit (Close App)".

Once the user click the OK button on the dialog box, the GUI will clear the quantity field for a new quantity to be entered, but leave the item number intact (see cropped image below).

This is a cropped screenshot of the same application window. The error dialog box is no longer present. The "Enter quantity for Item #2:" field is now empty, while the "Enter item ID for Item #2:" field still contains "8543". The "Details for Item #1:" and "Current Subtotal for 1 item(s):" fields remain unchanged, showing the same item details and \$17.98 subtotal.

17. The transaction file after several orders have been completed. Note the unique transaction ids based on the date and time for every separate transaction. This file shows seven separate transactions spanning different days.

```
eclipse-workspace-2024-06 - CNT 4714 - Project 1 - Fall 2024/transactions.csv - Eclipse IDE

SearchForItemInInventoryFile.java NileDotComFALL2024.java inventory.csv transactions.csv X
1 22082024200133, 14, "Stanley #2 Philips Screwdriver", 6.95, 1, 0.0, $6.95, August 22, 2024, 8:01:33 PM EDT
2 22082024200133, 339921, "Yamaha KT 100 kart engine", 950.0, 1, 0.0, $950.00, August 22, 2024, 8:01:33 PM EDT
3
4 23082024074752, 2190, "Hefty 30 gal with handles - box 40", 19.0, 2, 0.0, $38.00, August 23, 2024, 7:47:52 AM EDT
5 23082024074752, 298765, "3x5 index cards - 100 count", 2.25, 15, 0.2, $27.00, August 23, 2024, 7:47:52 AM EDT
6 23082024074752, 65435689, "12x16 paper cutter with ruler", 35.95, 1, 0.0, $35.95, August 23, 2024, 7:47:52 AM EDT
7 23082024074752, 2, "Tamiya 2oz paint - ferrari red", 2.39, 4, 0.0, $9.56, August 23, 2024, 7:47:52 AM EDT
8 23082024074752, 97634, "Campagnolo 12 spd crankset - 53/39 175mm", 750.0, 1, 0.0, $750.00, August 23, 2024, 7:47:52 AM EDT
9
10 23082024075127, 2, "Tamiya 2oz paint - ferrari red", 2.39, 1, 0.0, $2.39, August 23, 2024, 7:51:27 AM EDT
11
12 23082024075216, 234887, "2024 CSX train calendar", 12.95, 2, 0.0, $25.90, August 23, 2024, 7:52:16 AM EDT
13 23082024075216, RTX3452, "Laser Pointer", 38.8, 3, 0.0, $116.40, August 23, 2024, 7:52:16 AM EDT
14
15 23082024075409, 22345532, "3 ft mini USB cable M-F", 4.5, 25, 0.2, $90.00, August 23, 2024, 7:54:09 AM EDT
16 23082024075409, 14909, "1/4in - 28tpi SS nuts", 1.0, 50, 0.2, $40.00, August 23, 2024, 7:54:09 AM EDT
17 23082024075409, 8543, "Ritchey 1 inch threaded headset - silver", 45.0, 1, 0.0, $45.00, August 23, 2024, 7:54:09 AM EDT
18 23082024075409, 455654, "Park Tool 8mm threaded hook", 6.95, 30, 0.2, $166.80, August 23, 2024, 7:54:09 AM EDT
19 23082024075409, 115694, "Colnago C68 - UAE frame", 8500.0, 1, 0.0, $8,500.00, August 23, 2024, 7:54:09 AM EDT
20
21 23082024075450, 268, "Sidi Wire 2 carbon shoes - size 48", 499.99, 1, 0.0, $499.99, August 23, 2024, 7:54:50 AM EDT
22 23082024075450, 228865X, "3M Super 33+ electrical tape box 10", 32.5, 2, 0.0, $65.00, August 23, 2024, 7:54:50 AM EDT
23
24 23082024080226, 29741, "Plush teddy bear 8 inch", 24.0, 2, 0.0, $48.00, August 23, 2024, 8:02:26 AM EDT
25 23082024080226, 3451, "Oakley Jawbreaker sunglasses", 236.0, 2, 0.0, $472.00, August 23, 2024, 8:02:26 AM EDT
26 23082024080226, 22389, "Swingline Model B2200 Red stapler", 18.0, 1, 0.0, $18.00, August 23, 2024, 8:02:26 AM EDT
27 23082024080226, 198, "Box of 10 Sharpie Roller 0.5mm", 6.5, 5, 0.1, $29.25, August 23, 2024, 8:02:26 AM EDT
28
29 |
```